example, a Weiss AG Civetta spherical camera can record images with a dimension of 230 M pixels as it uses a rotation lens to scan surrounding scenes. This camera, however, can only be installed on a ground-fixed tripod since it consumes 40 seconds for each camera exposure.

- **The quality of reconstructed models.** The consumer-grade cameras usually consist of two or more dioptric camera modules, such as the Theta X camera with two lenses. This design model decreases the time costs consumed in outdoor data acquisition. However, they sacrifice the spatial resolution of recorded images, which can further reduce the quality of reconstructed models.

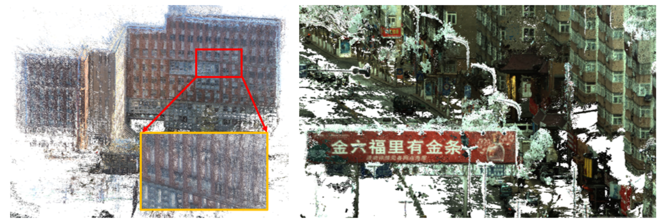
In order to increase the spatial resolution of spherical images, two possible solutions can be implemented without sacrificing the efficiency of data acquisition. On the one hand, more cameras can be integrated into one spherical camera as the spherical image is the fusion of images from these camera modules. This strategy has been widely used in recent spherical cameras, such as the Ladybug 5+ with six cameras and the Panono with 36 cameras; on the other hand, high-speed imaging could be a promising technology. It does not increase the size and weight of spherical cameras, and it decreases the time costs for recording high-resolution images [151].

C. Camera distortion and calibration

Serious geometric distortions exist in spherical images. On the one hand, the individual camera module has imaging geometric distortions; on the other hand, generating the spherical image would also introduce distortions. Especially for consumer-grade cameras, such as Ricoh Theta X and Insta360 Sphere, these cameras project multiple images from each camera onto a sphere and generate the ERP format spherical images. Generally, an ideal unit sphere camera model is used in the generation of the ERP format, as illustrated in Fig. 13(c), including the well-known commercial and open-source software packages Agisoft Metashape, Pix4Dmapper, and OpenMVG. This projection would introduce distortions and cause errors in 3D reconstruction. Fig. 25 presents a comparison of 3D models reconstructed from consumer-grade and professional spherical images. It is clearly shown that the point clouds in Fig. 25(a) are very coarse, even on the plain wall. On the contrary, the point clouds generated using the professional camera are more accurate, from which the text is very clear. Therefore, camera calibration should be seriously considered and executed for spherical images to improve the quality of reconstructed models.

D. Integration of aerial-ground image

Although spherical cameras can record 360 by 180 degrees of surrounding environments, their effective observation range is still limited to the near-ground region. In order to obtain complete reconstruction, aerial images must be integrated with ground spherical images. In the literature, there are valuable attempts for the integration of aerial-ground images in the context of 3D reconstruction, which can be roughly divided into 2D image-based methods and 3D point cloud-based methods [152].



(a) (b)

Fig. 25. The illustration of dense matching point clouds from (a) a consumer-grade camera; and (b) a professional camera.

- **2D image-based methods.** These methods aim to establish reliable correspondence between aerial and ground images and implement combined bundle adjustment by using both aerial and ground images. The commonly used strategies are image rectification and virtual rendering. The former aims to rectify aerial and ground images onto the same plane in the object space [153]; the latter utilizes the rough POS (Positioning and Orientation System) data of ground images and render virtual images from the 3D models reconstructed from aerial images [154]. The core idea of these methods is to decrease the geometric distortion caused by large viewpoints. However, for spherical images, their camera imaging model differs from the conventional perspective model. Thus, further consideration should be paid.
- **3D point cloud-based methods.** The core idea is to create 3D models separately from aerial and ground images and find reliable enough common 3D correspondences from these 3D models. The integration problem is then converted into the registration of 3D point clouds [152]. However, for spherical images, two issues should be considered. On the one hand, the trajectory of spherical images is limited by the structure of urban streets, which causes large accumulate drift in SfM or SLAM-based image orientation. Thus, it may be hard to model the registration between 3D point clouds by using the commonly used similarity transformation; on the other hand, as shown in Fig. 25, the reconstructed models of spherical images may contain many false 3D points due to camera distortions. Finding reliable correspondences becomes a non-trivial task between 3D point clouds created from varying images.

For the first issue, the recent deep learning-based technique can be used to achieve reliable feature detection and matching. Especially for decreasing the projection distortions, the DCN (deformable convolutional network) [155] based network can be a promising technique to achieve viewpoint invariant feature detection, and the GCN (graph convolutional network) based network, e.g., SuperGlue [156], can exploit the context information to execute reliable feature matching. For the second issue, the large image orientation problem can be divided into some clusters based on the street structures, and the accumulated drift can be decreased. This divide-and-conquer strategy has been used for efficient and accurate 3D reconstruction of UAV images [157].

V. CONCLUSIONS

Spherical images can record all surrounding environments by using one camera exposure. In contrast to perspective images with limited FOV, spherical images can cover the whole scene and have been increasingly used for 3D modeling in street-view and indoor environments. This paper reviews the 3D reconstruction from spherical images in terms of data acquisition, image matching, image orientation, and dense matching to wrap up recent techniques of 3D modeling based on spherical images. It also presents promising 3D reconstruction applications, including cultural heritage documentation, urban modeling and navigation, tunnel mapping and inspection, and other applications, e.g., urban tree localization, emergency response and rescue, and underwater collision avoidance. Finally, the main prospects are discussed in terms of the exploitation of crowdsource images, the spatial resolution of spherical images, camera distortion and calibration, and integration of aerial-ground images. According to this review, we can conclude that spherical images have great potential in the 3D reconstruction of street view and indoor scenes, and contemporary techniques can support their applications. Future studies can increase the spatial resolution of spherical cameras for data acquisition and exploit current deep learning-based methods to optimize the 3D reconstruction workflow.

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